

WASP-39b

R-1414

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-39_200527 · created 2026-08-01T05:32:53+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458996.7995 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1777 +/- 0.0094
Transit depth (Rp/Rs)^2	3.16 +/- 0.34 [%]
Orbital Inclination (inc)	86.09 +/- 0.19
Airmass coefficient 1 (a1)	0.965 +/- 0.013
Airmass coefficient 2 (a2)	0.0266 +/- 0.0094
Scatter in the residuals of the lightcurve fit is	1.35 %
Best Comparison Star	#3 - [450, 333]
Optimal Aperture	3.18
Optimal Annulus	8.56
Transit Duration (day)	0.1006 +/- 0.0041

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1777 $\pm$ 0.0094 vs 0.1457 (deviation 3.08 σ)
Duration fit vs published	0.1006 d vs 0.12 d (deviation 4.27 σ)
Mid-time O-C	32.3 min
Depth SNR	17.1
Residual scatter	1.35 %

Light curve

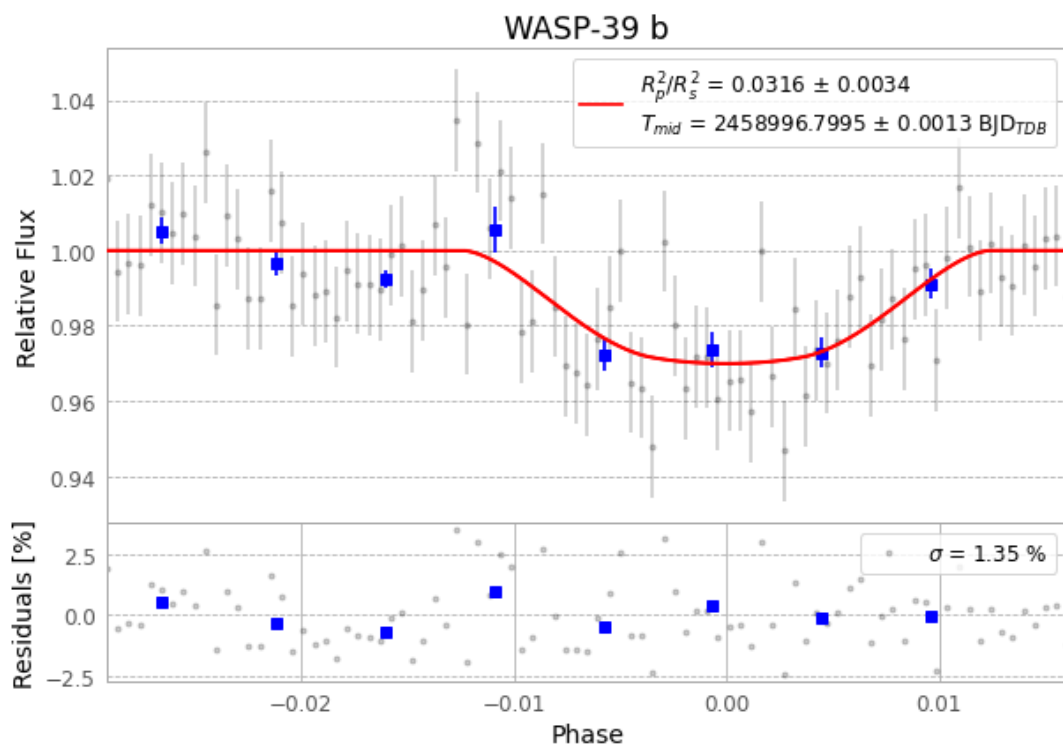


Figure 1 — FinalLightCurve_WASP-39 b_2020-05-26.png (SHA-256 daa5734be47fa126...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [201, 150], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9021dac0b818f4f2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

89 FITS frames (58 MB), WASP-39200527041212.FITS → WASP-39200527083612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-200527.log (d6909a9928d4...), FinalLightCurve_WASP-39 b_2020-05-26.png (daa5734be47f...), FinalLightCurve_WASP-39 b_2020-05-26.csv (6ba7f9f26a05...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 05:32Z.